January 2026

Dear Participants,

Welcome to the 2026 Higher Education Analytics Data Competition.

Please find included in this package information on the problem to be addressed and how to acquire datasets for Phase One of the competition. For instructions and details on competition rules and how to submit posters for the competition, please refer to the competition website (**https://datacompetition.mohawkcollege.ca**). Links to reference templates for submissions are available on the website as well. An electronic poster submission size of 36” L x 48” W (landscape) is required. These will be the sizes used if we print out physical copies of your poster.

Please remember that this is a student competition. Mentors are available to provide guidance and recommendations, but all analyses should be performed by student participants.

For Phase Two of the competition, qualifying teams will be provided with more details as the date approaches.

Best of luck to all the teams!

Sincerely,
HEAD Competition Organizing Committee

PROBLEM – PHASE ONE

TITLE: UNDERSTANDING THE CLIMATE CHANGE IMPACTS ON THE CITY'S TREE CANOPY

BACKGROUND

Project Statement: This is a case study involving the analysis of the City's tree inventory for students to apply data analytics to understand climate change impacts on the City's urban forest.

The City of Hamilton's Urban Forest Strategy is a high-level plan and the roadmap for a sustainable urban forest, and it sets the long-term direction for the City to manage this asset for the next 20 years. The Urban Forest Strategy includes guiding principles and actions that will assist the City in achieving its urban forest vision. One of the guiding principles is that trees are a valued city asset and an essential part of Hamilton's infrastructure, and the City is focused on actions that support the long term health of the urban forest. The urban tree canopy plays an important role in the City's Climate Change Impact Adaptation plan. This is reflected in Objective #8, which directs the City to "monitor, maintain, and improve the diversity and resiliency of urban trees and forests". It is also supported by the City's Urban Forest Strategy, specifically Action #17, which directs the City to "complete a climate change vulnerability assessment for Hamilton's natural systems, including the urban forest."

The urban forest is part of the green infrastructure in cities that assists with mitigating the impacts of climate change, including absorbing carbon dioxide, reducing major storm flooding, and providing shade to reduce energy demands. Although trees are an integral component of climate mitigation, tree canopies are also subject to climate change impacts, including extreme weather, drought, and intensified heat. Vulnerability assessments look at the local context and work with community input to prioritize and find the best ways to mitigate the risks and reduce the residual effects of climate change on Hamilton's urban forest.

The goal of this project is to analyze various data sources to demonstrate how climate change may impact the City's urban forest which includes evaluating vulnerability to extreme weather, droughts, assessing potential damage from storms or floods, and understanding the long-term effects of changing environmental conditions.

DESCRIPTION

Using the City's tree inventory and additional climate change data sets, students shall analyze how susceptible the City's urban tree canopy is to harm from climate impacts by reviewing risks and adaptive capacity in order to find ways to prioritize methods to build resilience. Students shall consider a wide range of factors, including tree species, tree size, geographical location, local climate modelling, and risk assessments, to provide suggestions for future growth and maintenance strategies for the urban forest for the short, medium, and long-term periods.

Possible areas to explore include:

What are the predominant climate risks facing the City's urban forest, and how will these risks impact the tree canopy over the short, medium, and long terms, i.e., by 2030, 2050, and 2080? Which types of trees and which areas within the City of Hamilton are most vulnerable to climate change risks? How can the City of Hamilton use its tree inventory and other tree health data to establish a framework to support the Urban Forest Strategy while mitigating climate change impacts for City-owned trees?

Datasets for this problem, in addition to any Open Data links to assist with your analysis, will be posted to a password-protected webpage. This webpage will be updated throughout the competition as more data and links are made available. Do NOT assume that the provided data is in a ready-to-use format. Some data may require preparation such as cleaning or transformation, before being analyzed.

2026 Data Link:

All core **Datasets** required for this project can be accessed through the City of Hamilton's Open Data Portal:

https://open.hamilton.ca/datasets/SpatialSolutions::public-tree-inventory/explore?showTable=true

Do not share this problem outside of your teammates. Sharing and distributing this problem information is expressly prohibited.

More information may be provided by the Competition Committee at a later date.
If there are any questions, you may submit them to datacompetition@mohawkcollege.ca.